
From Free to Fee

How Emission Permit Allocation Affects Firms

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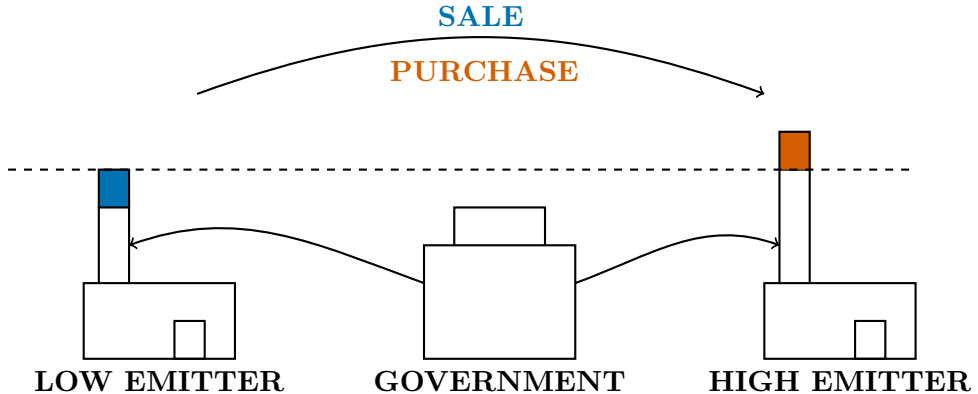
WCERE 2026

June 2026

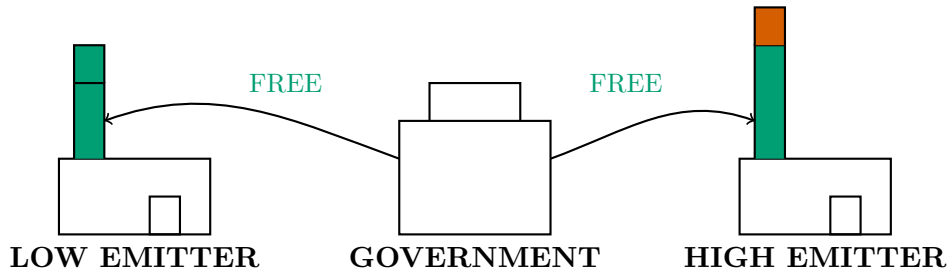
Motivation

- Share of ETS covered emissions increased to almost 20% (World Bank, 2025)
- Policy makers rely heavily on the **free emission permits** (Grubb et al., 2022)
- **Benefit:** Decrease cost and protection from competition (Martin et al., 2014)
- **Cost:** May also distort firms' emission responses (Hahn and Stavins, 2011)
- *Independence Property:*
Initial allocation does **not** affect firms' emission outcomes (Montgomery, 1972)
- In this paper:
 1. **Empirical evidence:** Fewer permits decrease emissions and economic activity
 2. **Consistent mechanism:** Multi-product general equilibrium model

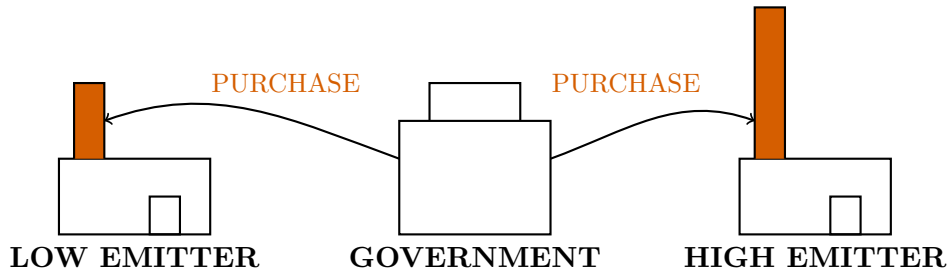
Emission Trading System



Allocation Mechanism - Free



Allocation Mechanism - Selling



The Paper

Question

What is the effect of decreasing free emission permit allocation on firms?

Setting and Methods

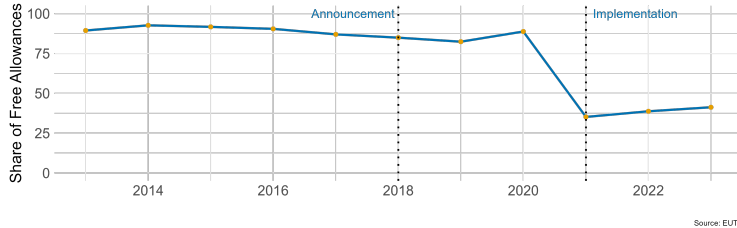
- Policy change in the allocation rule of free permits in EU ETS
- Manufacturing firm-level panel data (2014-2022) for DiD estimation
- Multi-product GE model based on Melitz (2003)

Results

1. **Empirics:** Substantial decrease of emissions and economic activity of firms
2. **Model:** Fewer free permits trigger the exit of the least productive products

The Reform

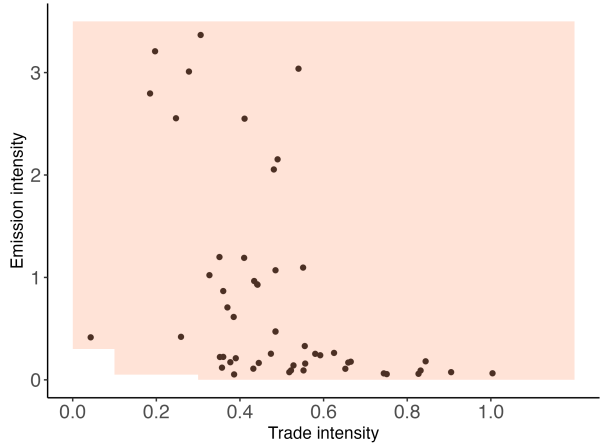
- This paper: focus on the **EU ETS**
- **Reform:** Some firms lost $\approx 50\%$ of their free emission permits
 - **2018:** Official publication of the list of sectors losing free allowances
 - **2021:** Implementation of the reform



Identification

Before Reform:

1. Emission intensity $\geq 5\%$ & Trade intensity $> 10\%$
2. Emission intensity $\geq 30\%$
3. Trade intensity $> 30\%$



► Data ► Calculation CI and TI ► Allocation Rule ► Detailed Timing

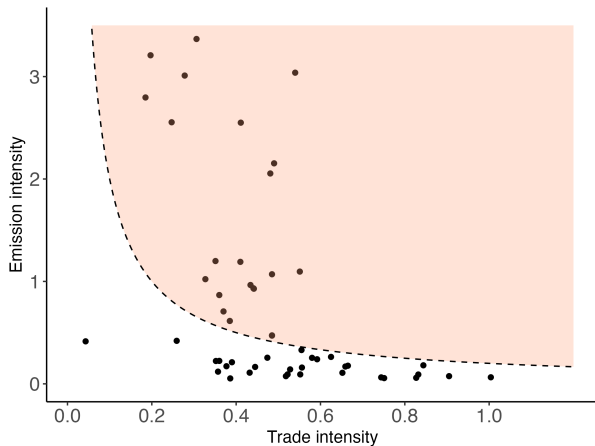
Identification

Before Reform:

1. Emission intensity $\geq 5\%$ & Trade intensity $> 10\%$
2. Emission intensity $\geq 30\%$
3. Trade intensity $> 30\%$

After Reform:

- Emission intensity $\times$ Trade intensity $> 20\%$



► Data

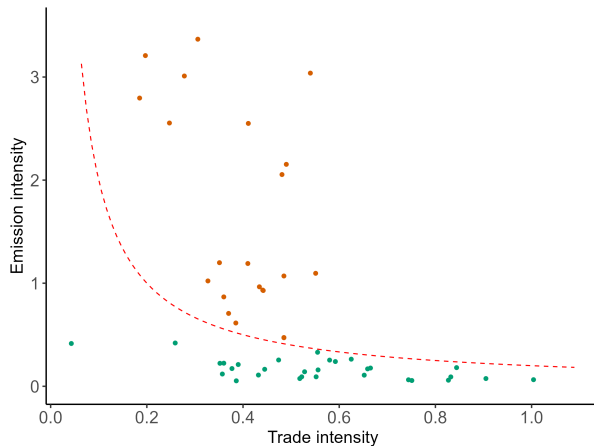
► Calculation CI and TI

► Allocation Rule

► Detailed Timing

Identification

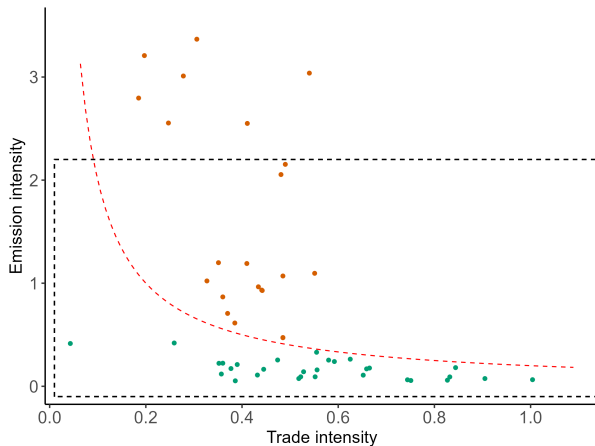
- Control group:
 - Before: **Free**
 - After: **Free**
- Treatment group:
 - Before: **Free**
 - After: **Pay**



► Data ► Calculation CI and TI ► Allocation Rule ► Detailed Timing

Identification

- Control group:
 - Before: **Free**
 - After: **Free**
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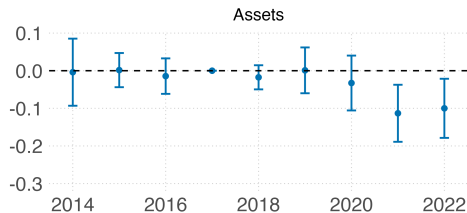
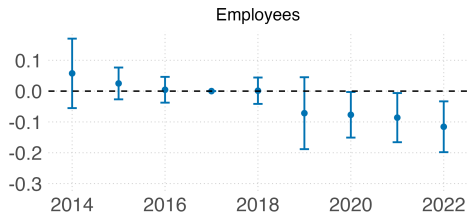
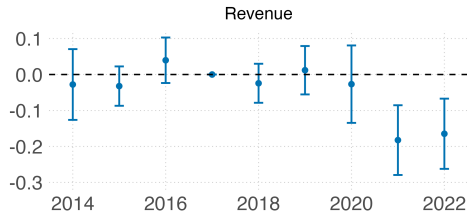
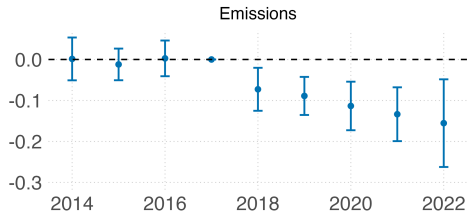
► Data

► Calculation CI and TI

► Allocation Rule

► Detailed Timing

Empirical Results



Model

- **Objective:**

How do free permits affect emissions and the economic activity of firms?

- **Environment:**

- General Equilibrium model based on Melitz (2003)
- Nested CES structure: Every firm j consists of many product lines
- Product lines differ only in their productivity φ
- Free allowances enter as a lump-sum transfer decreasing fixed cost

- **Key Mechanism:**

1. **Implementation:**

Free allowances $\downarrow$ fixed cost $\uparrow \Rightarrow$ Exit of low φ products $\Rightarrow$ firms shrink

2. **Announcement:**

Future revenue of surviving products $\uparrow \Rightarrow$ adopt technology

- Relative price index main object of interest:

$$\left(\frac{P_j}{P}\right)^{1-\varepsilon} = \frac{R_j}{R} = \frac{L_j^p}{L^p} = \frac{E_j^p}{E^p}, \quad \text{where} \quad \frac{P_j}{P} = \frac{M_j^{-\frac{1}{\sigma-1}} \tilde{\varphi}_j^{-1}}{\left(\sum_m M_m^{\frac{\varepsilon-1}{\sigma-1}} \tilde{\varphi}_m^{\varepsilon-1}\right)^{-\frac{1}{\varepsilon-1}}}$$

► Detailed Core

- Relative price index main object of interest:

$$\left(\frac{P_j}{P}\right)^{1-\varepsilon} = \frac{R_j}{R} = \frac{L_j^p}{L^p} = \frac{E_j^p}{E^p}, \quad \text{where} \quad \frac{P_j}{P} = \frac{M_j^{-\frac{1}{\sigma-1}} \tilde{\varphi}_j^{-1}}{\left(\sum_m M_m^{\frac{\varepsilon-1}{\sigma-1}} \tilde{\varphi}_m^{\varepsilon-1}\right)^{-\frac{1}{\varepsilon-1}}}$$

- M_j : total mass of product lines of firm j

► Detailed Core

Model

- Relative price index main object of interest:

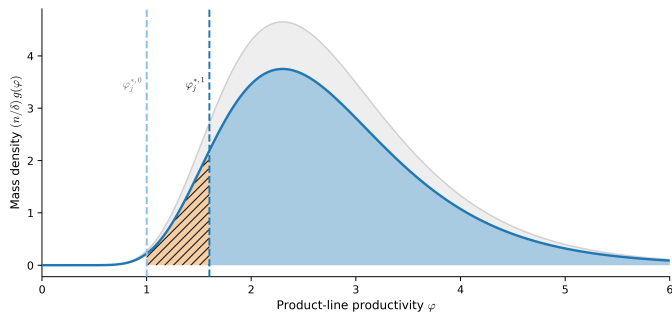
$$\left(\frac{P_j}{P}\right)^{1-\varepsilon} = \frac{R_j}{R} = \frac{L_j^p}{L^p} = \frac{E_j^p}{E^p}, \quad \text{where} \quad \frac{P_j}{P} = \frac{M_j^{-\frac{1}{\sigma-1}} \tilde{\varphi}_j^{-1}}{\left(\sum_m M_m^{\frac{\varepsilon-1}{\sigma-1}} \tilde{\varphi}_m^{\varepsilon-1}\right)^{-\frac{1}{\varepsilon-1}}}$$

- M_j : total mass of product lines of firm j
- $\tilde{\varphi}_j$: CES weighted average productivity of firm j

► Detailed Core

Policy: Implementation

$$\frac{\partial}{\partial F_j} \ln \left(\frac{P_j}{P_{-j}} \right) = \left[\underbrace{-\frac{1}{\sigma-1} \frac{1}{M_j} \frac{\partial M_j}{\partial F_j}}_{\text{selection and entry (+)}} - \underbrace{\frac{1}{\tilde{\varphi}_j} \frac{\partial \tilde{\varphi}_j}{\partial F_j}}_{\text{composition (-)}} \right] (+ \text{ GE spillovers})$$



$$x_j \downarrow \Rightarrow F_j \uparrow \Rightarrow M_j \downarrow \Rightarrow \frac{P_j}{P_{-j}} \uparrow$$

$$\Rightarrow \frac{R_j}{R_{-j}} \downarrow \Rightarrow \frac{L_j}{L_{-j}} \downarrow \Rightarrow \frac{E_j}{E_{-j}} \downarrow$$

Technology

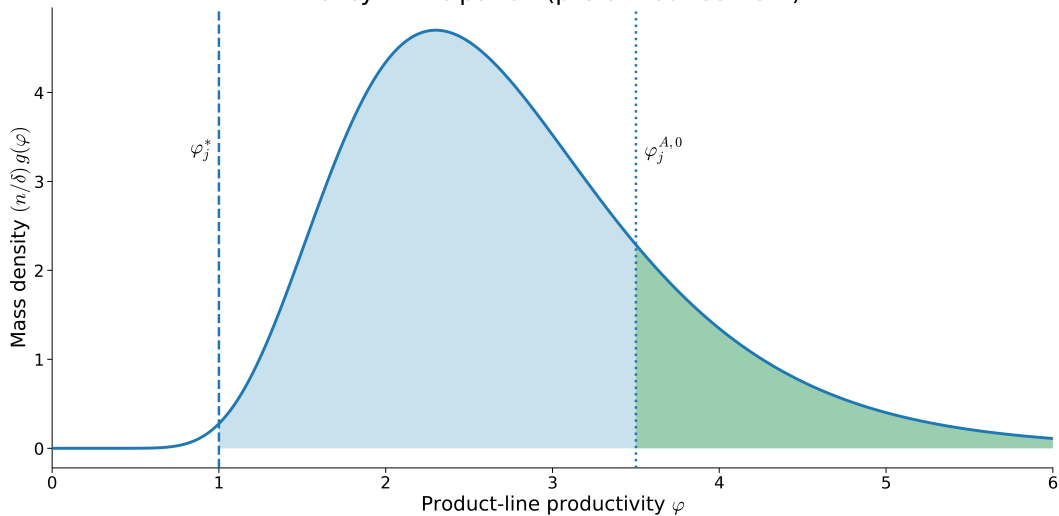
- Energy efficiency investment: $w\ell + \eta(P_e + \tau)e + F_j$
- A product line will adopt the technology iff $\frac{\Delta\pi(\varphi)}{\delta} \geq K$, which gives cutoff:

$$\varphi_j^A = \varphi_j^* \left[\frac{\sigma\delta K}{r_\ell(\varphi_j^*)(m^{1-\sigma} - 1)} \right]^{1/(\sigma-1)}, \quad \text{where} \quad m \equiv \left(\frac{\eta_h}{\eta_\ell} \right)^{1-\alpha}$$

$\Rightarrow$ A product line adopts the technology if its revenue is large enough

► Detailed Technology

Policy: Anticipation (pre-announcement)



Policy: Anticipation

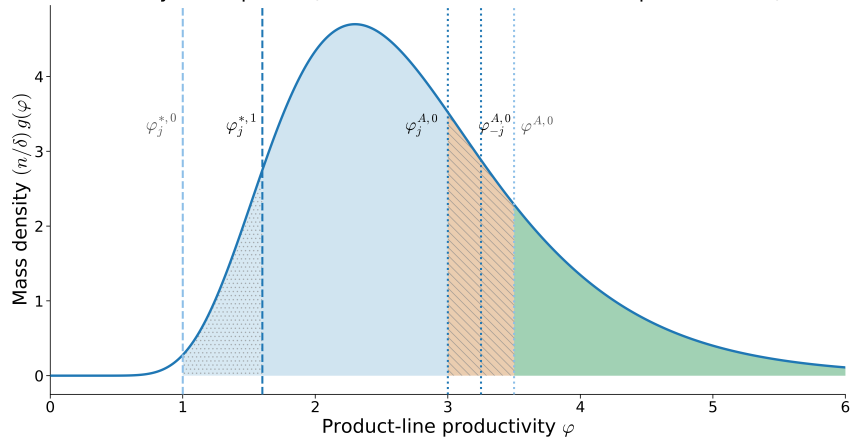
- **Remaining** product lines increase lifetime profits: $\Delta\pi_0(\varphi) + \frac{1-\delta}{\delta}\Delta\pi_1(\varphi)$
- Anticipation effect $\Phi > 1$ increases technology adoption:

$$\varphi_{A,j}^0 = \varphi_j^{*0} \left[\frac{\sigma\delta K}{r_\ell^0(\varphi_j^*)(m^{1-\sigma} - 1)} \cdot \frac{1}{1 + \frac{1-\delta}{\delta}\Phi} \right]^{\frac{1}{\sigma-1}}, \quad \Phi \equiv \frac{r_\ell^1(\varphi)}{r_\ell^0(\varphi)}$$

- Within firm j **more** product lines adopt the technology, because $\sigma > \varepsilon$

► Detailed Anticipation

Policy: Anticipation (after announcement, before implementation)



$$F_j^1 \uparrow \Rightarrow M_j^1 \downarrow \Rightarrow \frac{\varphi_{A,j}^0}{\varphi_{A,-j}^0} \downarrow \Rightarrow \frac{E_j^0/R_j^0}{E_{-j}^0/R_{-j}^0} \downarrow \Rightarrow \frac{E_j^0}{E_{-j}^0} \downarrow \left| \frac{R_j^0}{R_{-j}^0} \text{ fixed} \right.$$

Contribution

1. General equilibrium effects of permit allocation

- Montgomery (1972); Hahn and Stavins (2011); Fowlie et al. (2016); Konishi and Tarui (2015); Anouliès (2017); Dardati and Saygili (2020)

↔ This paper: Intensive margin through product adjustment

2. Empirical evidence on permit allocation

- (Fowlie and Perloff, 2013; Zaklan, 2023; Marchioro, 2026; Pacelli, 2026; Köveker and Sogalla, 2026)

↔ This paper: New source of variation and all European manufacturing

3. Broader literature on ETS and firm behavior

- Karp and Traeger (2025); Dechezleprêtre et al. (2023); Bayer and Aklin (2020); Martin et al. (2016); Colmer et al. (2025); Janser et al. (2025); Löschel et al. (2019); Ahlvik and Liski (2022)

↔ This paper: Focus on free emission permit allocation

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Appendix

Allocation Rule

Rule:

$$A_{ist} = B_s \cdot \text{HAL}_{i\bar{t}} \cdot R_t \cdot \mathbf{CLEF}_{st}$$

1. B_s : average allowances required by 10% most efficient installations in sector s
2. $\text{HAL}_{i\bar{t}}$: median annual activity level of installation i in reference period $\bar{t}$
3. R_t : ensures to not go beyond the overall cap at time t
4. $\mathbf{CLEF}_{st}$: if sector s is at "risk of carbon leakage", it is equal to 100%

► Back

Example Allocation Rule

For installation i in the steel sector:

- B_s : 1.8 allowances per ton of steel
- $HAL_{i\bar{t}}$: 200,000 tons of steel produced annually
- R_{it} : 0.9 (to align with the total EU ETS cap)
- $CLEF_{st}$: on the list (1.0) or not (0.3)

$$\begin{aligned}A_{ist} &= B_s \cdot HAL_{i\bar{t}} \cdot R_{it} \cdot CLEF_{st} \\&= 1.8 \cdot 200,000 \cdot 0.9 \cdot \textcolor{red}{1.0} \text{ (0.3)} \\&= 324,000 \text{ (97,200) allowances.}\end{aligned}$$

Exposure calculation

$$\text{Emission intensity} = \frac{(\text{direct emissions} \cdot \text{auctioning factor} + \text{indirect emissions}) \cdot 30\text{€}/\text{tonCO}_2}{\text{gross value added at factor costs}}$$

$$\text{Trade intensity} = \frac{\text{imports} + \text{exports}}{\text{imports} + \text{production}}$$

Example: Restriction on Emissions

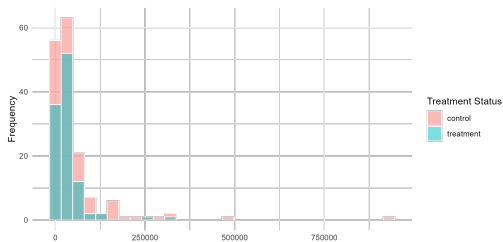


Figure: Distribution Before Restriction

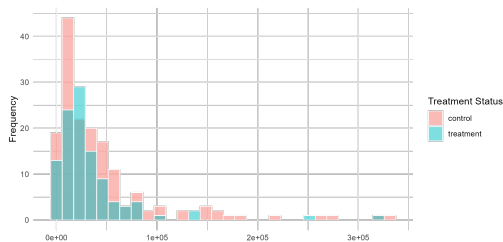
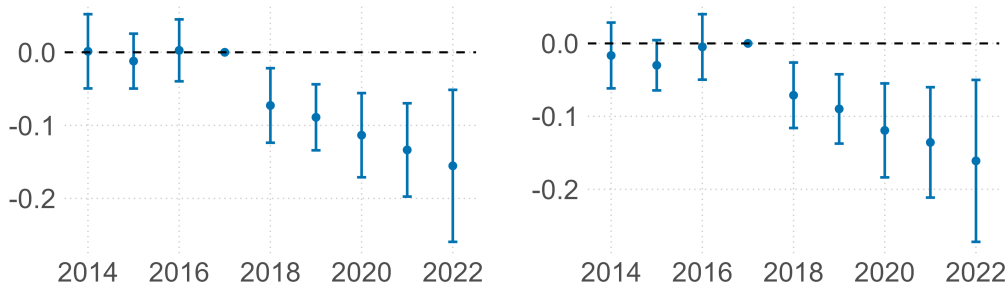


Figure: Distribution After Restriction

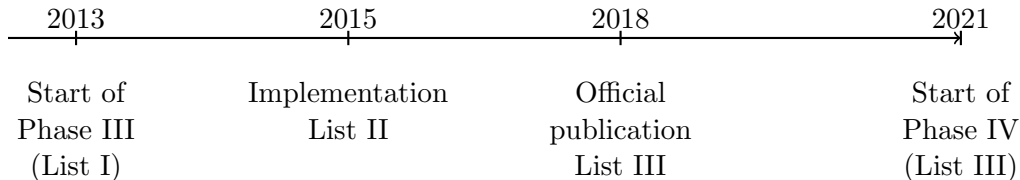
► Back

Restriction: Emission Result

Figure: Estimation without restriction (left panel) vs. with restriction (right panel)



Timing



▶ Back

Sample: Firm-by-year balanced panel from 2014 to 2022

1. **Emissions**
2. **Economic Performance**
3. **Treatment Status**

► Back

Sample: Firm-by-year balanced panel from 2014 to 2022

1. Emissions

- **Installation-by-year**
- Source: Jan Abrell, from EUTL
- Universe of regulated emissions under EU ETS
- Sector at NACE 4 and Orbis identifier

2. Economic Performance

3. Treatment Status

Sample: Firm-by-year balanced panel from 2014 to 2022

1. Emissions

2. Economic Performance

- Firm-by-year
- Source: Orbis Europe database
- Business information on Europe's largest 600,000 companies
- We extract turnover, total assets, and employees

3. Treatment Status

Sample: Firm-by-year balanced panel from 2014 to 2022

1. Emissions

2. Economic Performance

3. Treatment Status

- **Sector-before-after**
- Source: European Commission
- Detailed documentation on carbon leakage status (treatment)
- Variation on the NACE 4 level
- Official Emission intensity and Trade intensity

Data

	Control					Treatment				
	N	Mean	Min	Max	SD	N	Mean	Min	Max	SD
<i>Financial Variables</i>										
Revenue	2 246	580 901	389.68	18 750 395	1 230 604	884	2 495 986	3 548	92 654 000	9 031 196
Employees	2 246	944	1	104 023	3 676	884	8 097	4	413 811	38 656
Assets	2 246	502 534	290	11 383 277	1 100 323	884	3 434 308	5 716	126 285 000	13 641 825
<i>Emissions</i>										
Emissions	2 763	125 456	434	3 911 793	317 161	1 053	32 546	249	412 137	42 010
Free / Emissions (% , 2020)	307	121	0	664	98	116	106	0	760	85
Free / Emissions (% , 2021)	307	99	1	595	82	116	50	0	403	55

► Back

$$\ln(y_{i,s,t}) = \alpha_i + \gamma_t + \sum_{k=-3}^6 \beta_k (\mathbb{1}[t - c_i = k]) + \varepsilon_{i,s,t}$$

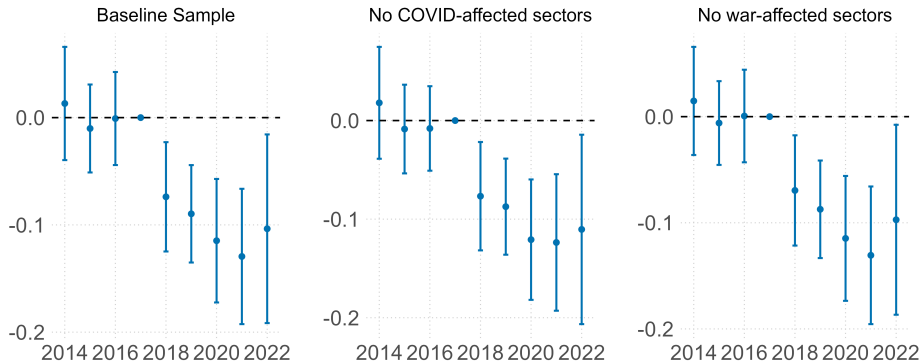
- $y_{i,s,t}$ is the outcome of interest for firm i in sector s in time t
- α_i and γ_t are firm and time fixed effects
- Standard errors are clustered at the sector level
- $c_i \in \{-\infty, 2018\}$ is equal to 2018 if the firm is treated, $-\infty$ otherwise

► Back

Main Result: Emissions

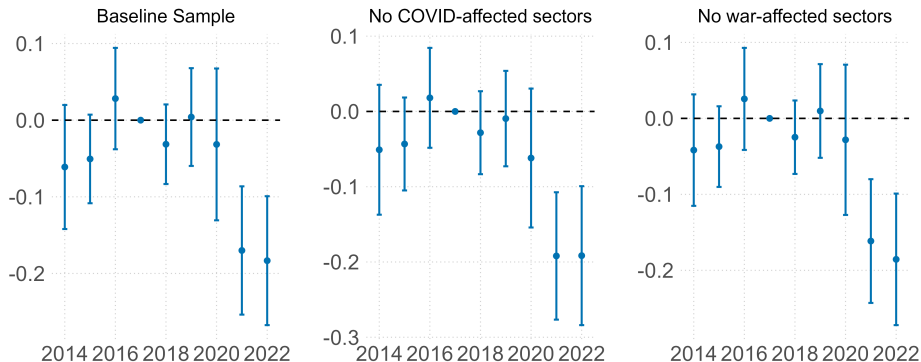
Dependent Variables:	Verified Emissions	Revenue	Employment	Assets
<u>Variables</u>				
Treatment $\times$ Post (2018)	-0.090*** (0.026)	-0.012 (0.038)	-0.066* (0.038)	-0.012 (0.029)
Treatment $\times$ Post (2021)	-0.053** (0.025)	-0.161*** (0.032)	-0.051** (0.022)	-0.091*** (0.022)
<u>Fixed-effects</u>				
Firm FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
<u>Fit statistics</u>				
Observations	3,816	3,124	3,124	3,124
R ²	0.97649	0.97554	0.97422	0.98312

COVID and the Ukraine War: Emissions



- **COVID-affected sector:** Pharmaceutical manufacturing (*C2120*)
- **War-affected sectors:** Weapons and shipbuilding (*C2540*, *C3011*)

COVID and the Ukraine War: Turnover



- **COVID-affected sector:** Pharmaceutical manufacturing (*C2120*)
- **War-affected sectors:** Weapons and shipbuilding (*C2540*, *C3011*)

Matching

- Methodology commonly used (Colmer et al., 2025; Zaklan, 2023)
- One-to-one nearest-neighbor matching using a mahalanobis distance
- Firms are matched based on emission intensity and total assets
- Results remain robust using:
 - No replacement
 - Euclidean distance
 - Up to 4 nearest neighbors

Matching Results

	Verified Emissions	Revenue	Employees	Assets
Treatment 2018	-0.101*** (0.026)	0.005 (0.024)	-0.017 (0.025)	0.005 (0.023)
Treatment 2021	-0.111*** (0.036)	-0.086** (0.038)	-0.005 (0.028)	-0.081** (0.036)
Observations	3025	2863	2863	2863
Adjusted R^2	0.032	0.009	-0.000	0.010

Alternative Matching Specification

	Euclidean	No replacement	4NN	3NN	2NN	1NN
Treatment 2018	-0.104*** (0.028)	-0.107*** (0.025)	-0.098*** (0.025)	-0.103*** (0.027)	-0.108*** (0.025)	-0.106*** (0.032)
Treatment 2021	-0.100*** (0.037)	-0.100*** (0.034)	-0.112*** (0.036)	-0.113*** (0.038)	-0.123*** (0.036)	-0.113** (0.045)
Observations	3025	3025	2420	1815	1210	605
Adjusted R^2	0.027	0.029	0.032	0.032	0.039	0.031

► Back

Industry Trends

Dependent Variables: Model:	Verified Emissions (1)	Revenue (2)	Employees (3)	Assets (4)
<u>Variables</u>				
Treatment $\times$ Post (2018)	-0.0681** (0.0293)	-0.0400 (0.0350)	0.0177 (0.0395)	-0.0024 (0.0198)
Treatment $\times$ Post (2021)	-0.0374 (0.0273)	-0.1856*** (0.0447)	0.0098 (0.0369)	-0.0803*** (0.0227)
<u>Fixed-effects</u>				
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
<u>Varying Slopes</u>				
Industry	Yes	Yes	Yes	Yes
<u>Fit statistics</u>				
Observations	3,816	3,124	3,124	3,124
R ²	0.97776	0.97711	0.97617	0.98500
Within R ²	0.00114	0.00802	5.8×10^{-5}	0.00263

Country Specific Trends

Dependent Variables: Model:	Verified Emissions (1)	Revenue (2)	Employees (3)	Assets (4)
<u>Variables</u>				
Treatment x Post (2018)	-0.0828*** (0.0278)	-0.0040 (0.0430)	-0.0498 (0.0358)	-0.0031 (0.0295)
Treatment x Post (2021)	-0.0560** (0.0240)	-0.1825*** (0.0313)	-0.0486** (0.0211)	-0.0946*** (0.0227)
<u>Fixed-effects</u>				
Firm FE	Yes	Yes	Yes	Yes
Country x Time FE	Yes	Yes	Yes	Yes
<u>Fit statistics</u>				
Observations	3,789	3,094	3,094	3,094
R ²	0.97770	0.97674	0.97666	0.98400
Within R ²	0.01330	0.01681	0.00441	0.00675

EUA prices and the Reform

- EUA prices may affect empirical results in two ways:
 1. Value of free allowances → amplifying treatment
 2. Direct effect of EUA prices on firms' emissions
- Different effect in the groups either through exposure or abatement elasticity:

$$\Delta CO_2 = \underbrace{\delta}_{\text{abatement elasticity } (\leq 0)} \cdot \underbrace{E^0}_{\text{baseline exposure}} \cdot \underbrace{\Delta P}_{>0}$$

- **Exposure:** Higher in the control group → estimate lower bound
- **Abatement:** No differential effect of prices pre-treatment

- **Data:** ComExt Intra- and Extra-European trade database (sector level)
- **Empirical Strategy:**

1. **Standard:**

$$\ln y_{ji,s,t} = \tau_{ijt} + \tau_{ijs} + \rho D_{st} + \alpha T_{ijst} + u_{ji,s,t}$$

2. **Triple difference:**

$$\ln y_{ji,s,t} = \tau_{ijt} + \tau_{ijs} + \delta_{st} + \rho_{DDD} [D_{st} \times ExtraETS_j] + \alpha T_{ijs} + \varepsilon_{ijst}$$

⇒ DDD allows for controlling on sector $\times$ time level

Trade

	Standard		Triple	
Dependent Variables:	Exports (logs)	Imports (logs)	Exports (logs)	Imports (logs)
<u>Variables</u>				
Treatment Effect 2018	-0.0328 (0.0380)	-0.0114 (0.0427)	-0.0611* (0.0342)	-0.0177 (0.0697)
<u>Fixed-effects</u>				
Sector-i-j	Yes	Yes	Yes	Yes
Time-i-j	Yes	Yes	Yes	Yes
Sector-Time			Yes	Yes
<u>Fit statistics</u>				
Observations	331,166	382,558	331,166	382,558
R ²	0.92477	0.91535	0.92554	0.91596
Within R ²	5.43×10^{-5}	4.26×10^{-6}	4.07×10^{-5}	2.22×10^{-6}

Clustered (Sector) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Demand

- C.E.S. utility function over a continuum of goods:

$$U = \left(\sum_{j=1}^N Q_j^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}} \quad \text{where} \quad Q_j = \left(\int_{\Omega_j} q_j(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \quad .$$

- Ω_j is the mass of goods of firm j indexed by ω
- Set of varieties can be understood as an aggregated good $Q = U$ with price P .
- Elasticity of substitution larger within than across firms: $\sigma > \varepsilon > 1$

Production

- Production of a **product line** requires labor and energy
- Each product line draws a productivity level φ_j from $g(\varphi)$
- Lump-sum free allowances ($\mathbf{x}_j$) enter fixed cost with permit price τ

$$\min_{\ell, e} w\ell + (P_e + \tau)e + \underbrace{f_{fix} - \mathbf{x}_j\tau}_{F_j} \quad \text{s.t.} \quad \varphi\ell^\alpha e^{1-\alpha} = 1.$$

- Minimizing cost yields the **unit cost** $c_u(\varphi)$ and **pricing rule** $p_j(\varphi)$:

$$c_u(\varphi) = \frac{1}{\varphi} \underbrace{\left(\frac{w}{\alpha}\right)^\alpha \left(\frac{P_e + \tau}{1 - \alpha}\right)^{1-\alpha}}_{\equiv \Psi}, \quad p_j(\varphi) = \underbrace{\frac{\sigma}{\sigma - 1}}_{\text{constant markup}} \frac{\Psi}{\varphi}$$

$\Rightarrow$ **Pricing independent of F_j**

Aggregation

- The aggregate price of firm j is:

$$P_j = \underbrace{\left(\frac{\sigma}{\sigma-1} \frac{\Psi}{\tilde{\varphi}_j} \right)}_{p_j(\tilde{\varphi}_j)} M_j^{-\frac{1}{\sigma-1}}, \quad \text{where} \quad \tilde{\varphi} \equiv \left[\int_0^\infty \varphi^{\sigma-1} \mu(\varphi) d\varphi \right]^{-\frac{1}{\sigma-1}}.$$

- The relative price index is independent of input prices and markup:

$$\frac{P_j}{P} = \frac{M_j^{-\frac{1}{\sigma-1}} \tilde{\varphi}_j^{-1}}{\left(\sum_m M_m^{\frac{\varepsilon-1}{\sigma-1}} \tilde{\varphi}_m^{\varepsilon-1} \right)^{-\frac{1}{\varepsilon-1}}}$$

Policy: Implementation

1. **Entry-flow channel:** Higher F_j discourages entry $\frac{dM_j}{dF_j} < 0$
2. **Selection channel:** Higher F_j rise survival cutoff: $\frac{\partial M_j}{\partial F_j} < 0$
3. **Composition channel:** Remaining product lines more productive $\frac{\partial \tilde{\varphi}_j}{\partial F_j} > 0$

Implementation channel M_j

- M_j is given by stationarity:

$$\underbrace{\delta M_j}_{\text{Exit}} = \underbrace{n_j [1 - G(\varphi_j^*)]}_{\text{Entry}} \Rightarrow M_j = \frac{n_j}{\delta} [1 - G(\varphi_j^*)].$$

- Then the effect of F_j on M_j is:

$$\frac{dM_j}{dF_j} = \underbrace{\frac{dn_j}{dF_j} \frac{1 - G(\varphi_j^*)}{\delta}}_{\text{Entry-flow (-)}} - \underbrace{\frac{n_j}{\delta} g(\varphi_j^*) \frac{d\varphi_j^*}{dF_j}}_{\text{Selection (-)}} < 0$$

- n_j is the inflow of product lines and φ_j^* is the cutoff productivity
- We can show that $\frac{dn_j}{dF_j} < 0$ and $\frac{d\varphi_j^*}{dF_j} > 0$ using a pareto tail distribution

Implementation channel $\tilde{\varphi}_j$

- $\tilde{\varphi}_j$ is given by:

$$\tilde{\varphi}_j(\varphi_j^*) = \left[\frac{1}{1 - G(\varphi_j^*)} \int_{\varphi_j^*}^{\infty} \varphi_j^{\sigma-1} g(\varphi_j) d\varphi_j \right]^{-\frac{1}{\sigma-1}}.$$

- Then the effect of F_j on $\tilde{\varphi}_j$ is:

$$\frac{\partial \tilde{\varphi}_j}{\partial F_j} = \underbrace{\frac{d\tilde{\varphi}_j}{d\varphi_j^*}}_{\text{truncation raises mean (+)}} \underbrace{\frac{d\varphi_j^*}{dF_j}}_{\text{tougher cutoff (+)}} > 0$$

- We can show that $\frac{d\tilde{\varphi}_j}{d\varphi_j^*} > 0$ and $\frac{d\varphi_j^*}{dF_j} > 0$ using a Pareto tail distribution

Technology

- Introduction of technology for higher energy efficiency: $\eta_l > \eta_h$ for K :

$$\min_{\ell, e} w\ell + \eta(P_e + \tau)e + F_j \Rightarrow p(\eta; \varphi) = \frac{\sigma}{\sigma - 1} \frac{\Psi(\eta)}{\varphi},$$

- "High efficiency" technology increases profits by:

$$\Delta\pi(\varphi) = \frac{1}{\sigma}(m^{1-\sigma} - 1)r_\ell(\varphi), \quad \text{where} \quad m \equiv \frac{\Psi(\eta_h)}{\Psi(\eta_\ell)} = \left(\frac{\eta_h}{\eta_\ell}\right)^{1-\alpha}$$

- A product lines will adopt the technology iff $\frac{\Delta\pi(\varphi)}{\delta} \geq K$:

$$\varphi_j^A = \varphi_j^* \left[\frac{\sigma\delta K}{r_\ell(\varphi_j^*)(m^{1-\sigma} - 1)} \right]^{1/(\sigma-1)}.$$

Policy: Anticipation

- **Remaining** product lines increase lifetime profits:

$$\begin{aligned}\Delta\pi(\varphi) &= \Delta\pi_0(\varphi) + (1-\delta)\Delta\pi_1(\varphi) + (1-\delta)^2\Delta\pi_1(\varphi) + \dots \\ &= \Delta\pi_0(\varphi) + \frac{1-\delta}{\delta}\Delta\pi_1(\varphi)\end{aligned}$$

- Anticipation effect $\Phi > 1$ increases adoption:

$$\varphi_{A,j}^0 = \varphi_j^{*0} \left[\frac{\sigma\delta K}{r_\ell^0(\varphi_j^*)(m^{1-\sigma} - 1)} \cdot \frac{1}{1 + \frac{1-\delta}{\delta}\Phi} \right]^{\frac{1}{\sigma-1}}, \quad \Phi \equiv \frac{r_l^1(\varphi)}{r_l^0(\varphi)}$$

$$\left. \frac{\partial}{\partial F_j^1} \left(\frac{\varphi_{A,j}^0}{\varphi_{A,m}^0} \right) \right|_{\Phi=1} = -\frac{1-\delta}{\sigma-1}(\sigma - \varepsilon) \left(\underbrace{\frac{1}{P_j^1} \frac{\partial P_j^1}{\partial F_j^1}}_{+} - \underbrace{\frac{1}{P_m^1} \frac{\partial P_m^1}{\partial F_j^1}}_{+} \right)$$